IBM DATA ANALYST CAPSTONE PROJECT

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OUTLINE

- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion

EXECUTIVE SUMMARY

- **Current Technology Usage**
  - Top 10 Languages
  - Top 10 Databases
  - Top 10 Platforms
  - Top 10 Web frames
- **Future Technology Trends**
  - Top 10 Languages desired next year
  - Top 10 Databases desired next year
  - Top 10 Platforms desired next year
  - Top 10 Web frames desired next year
- **Demographics**
  - Respondent Distribution by Age
  - Respondent Count by Country
  - Respondent Distribution by Education level
  - Respondent Count by Age, classified by Education level

INTRODUCTION

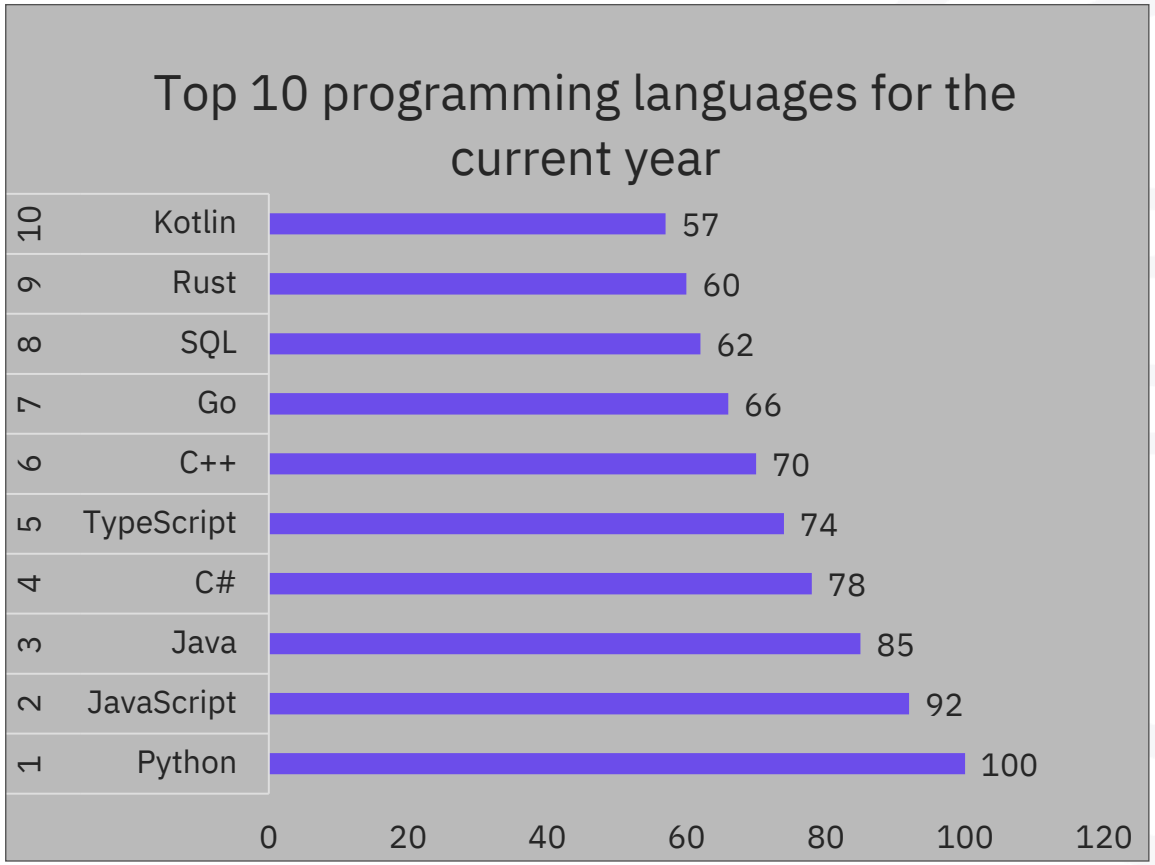
- Emerging Trends in Data Analysis
- AI and machine learning integration
- Improved data governance and ethics frameworks
- Real-time processing and streaming analytics
- Augmented analytics
- Cloud-native data analytics
- Enhanced data visualization and storytelling
- Advanced predictive and prescriptive analytics

METHODOLOGY

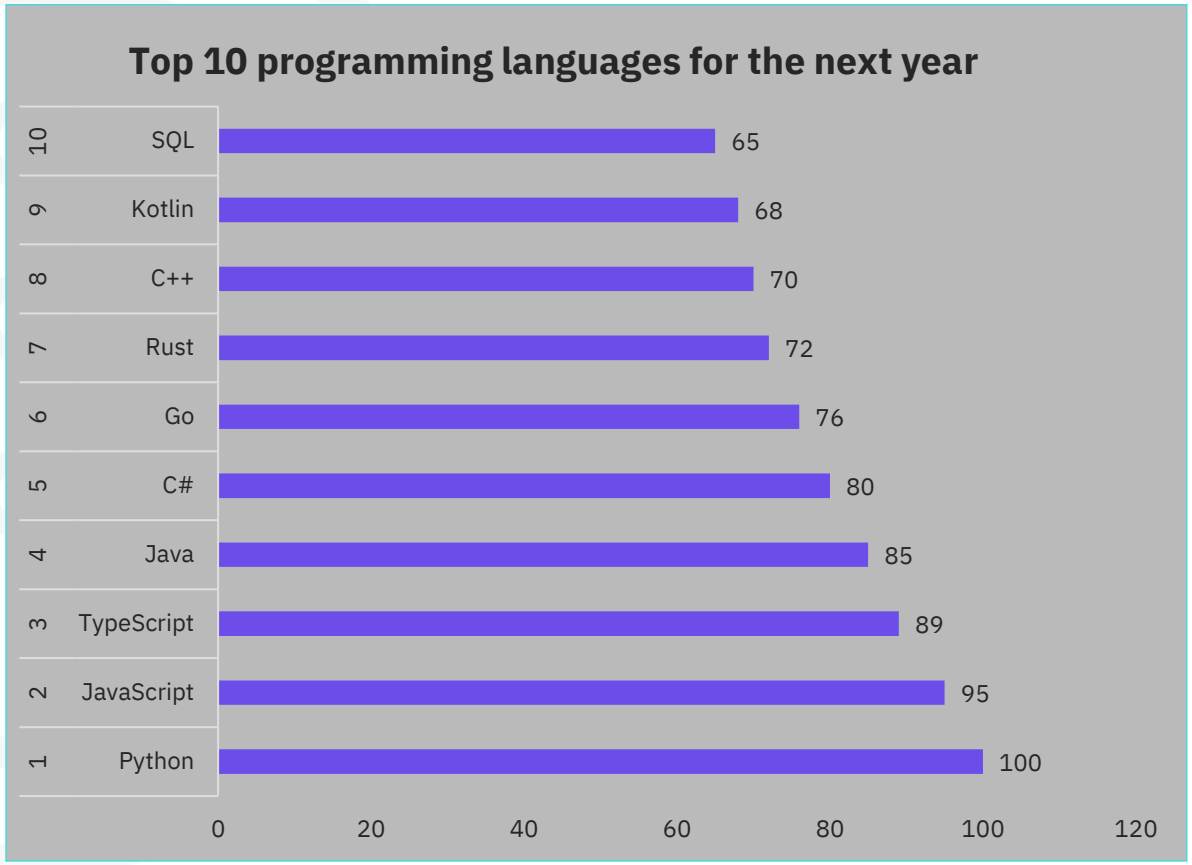
- Collecting and Exploring Data using APIs and Web Scraping
- Data Wrangling
- Performing Exploratory Data Analysis
- Data Visualisation
- Dashboard Building

PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

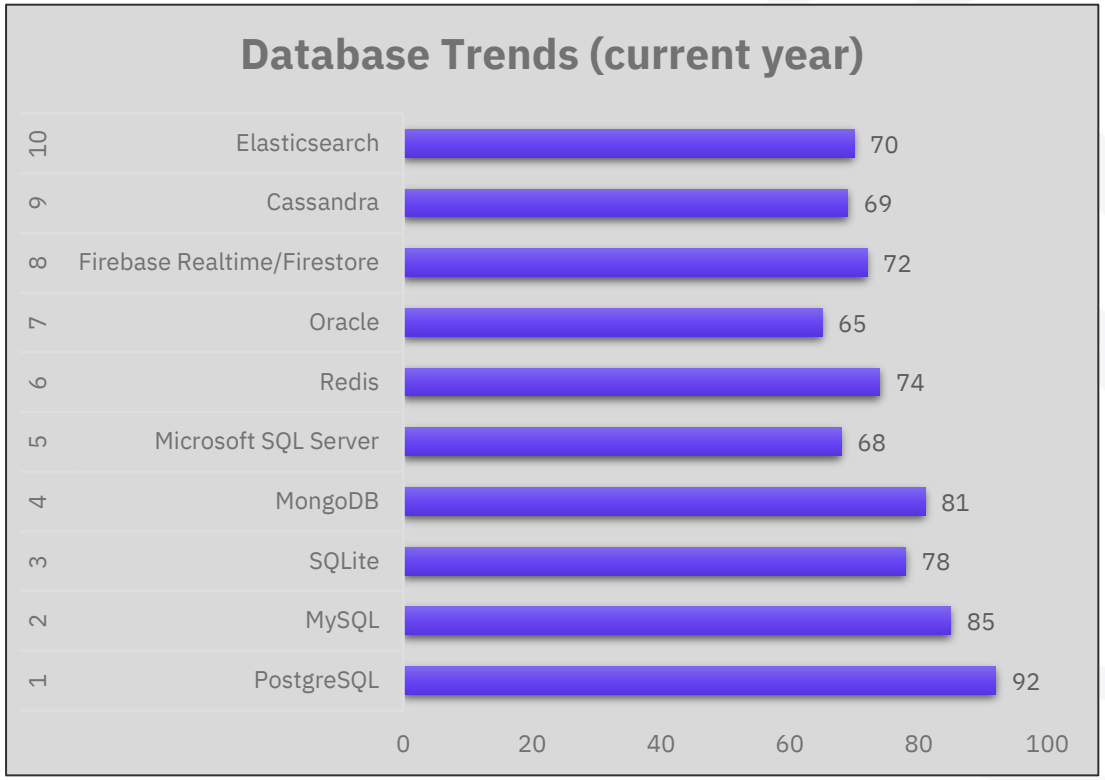
- Python and JavaScript are unshaken at the top
- TypeScript shows real growth reflecting its expanding use in enterprise-scale front-end development
- Rust gains momentum, overtaking C++ and approaching mid-tier positions
- Go is rising, due to demand for fast, efficient back-end and cloud-native applications
- SQL and PHP are declining slowly

Implications

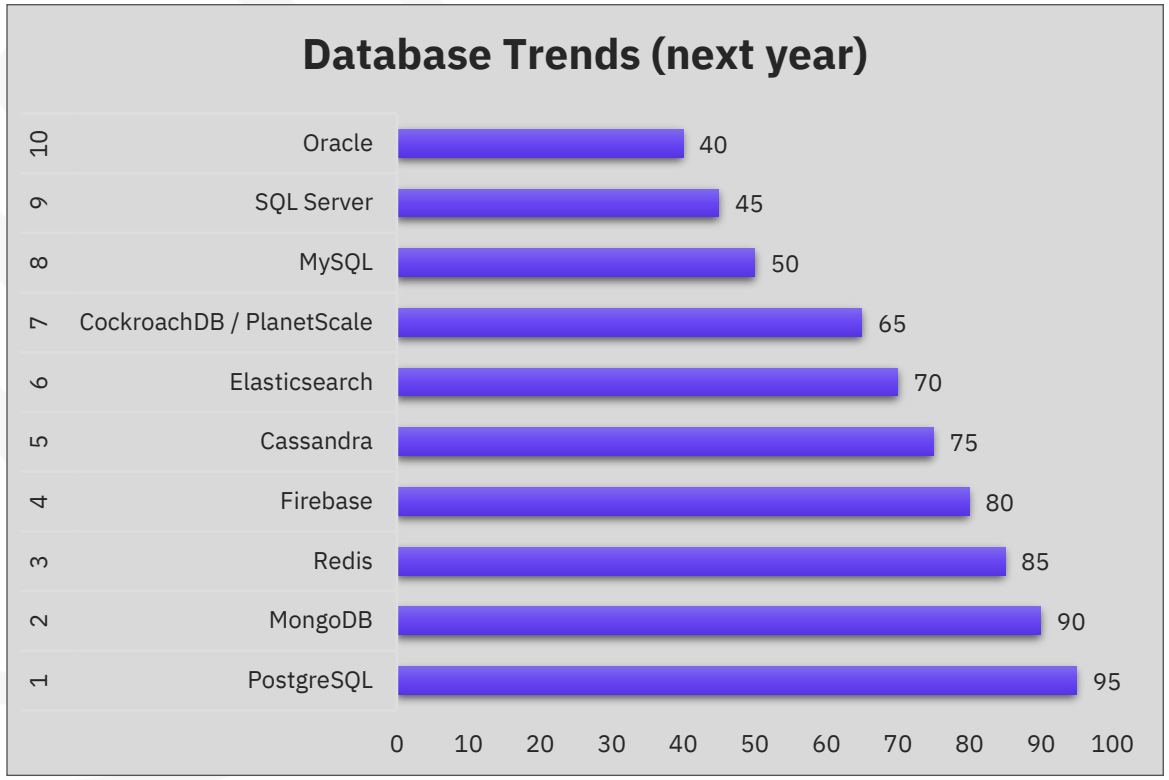
- Learning TypeScript and Rust is a smart investment
- Web developers should prioritize JavaScript and TypeScript
- Projects in legacy languages like PHP or C++ may face talent shortages as fewer new developers learn them
- Hiring in Python and TypeScript will remain competitive
- Modernizing back-end infrastructure with Go or Rust could improve performance and reduce long-term technical debt

DATABASE TRENDS

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is dominating
- MongoDB's NoSQL leadership is growing
- Redis is critical for real-time applications
- Firebase adoption is startup-driven
- Cassandra remains strong in big data and IoT
- Traditional RDBMS (MySQL, SQL Server, Oracle) are in slow decline
- Emerging players like CockroachDB and PlanetScale are gaining attention

Implications

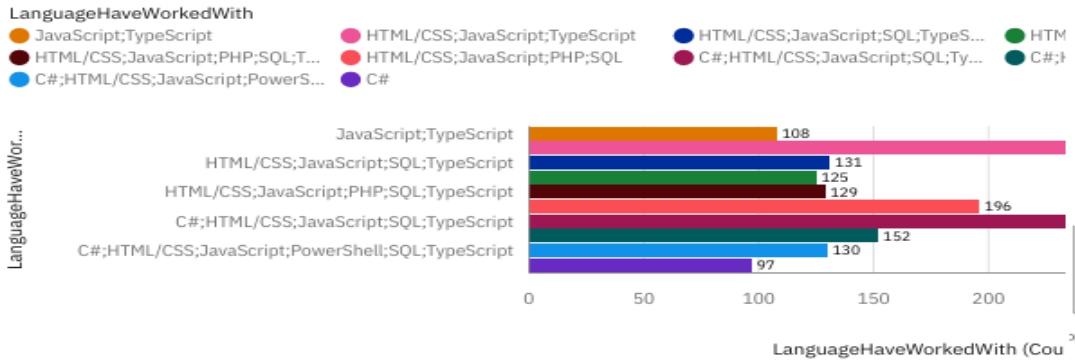
- Invest in PostgreSQL and MongoDB skills — they'll continue to dominate hiring trends and modern architecture stacks
- Understand Redis deeply — especially for roles involving real-time systems, gaming, or ML integration
- Reevaluate Oracle and SQL Server costs — as scalable and open-source alternatives mature
- Adopt developer-friendly, serverless databases for new services (like PlanetScale or CockroachDB) to enable fast scaling without deep infra overhead

DASHBOARD

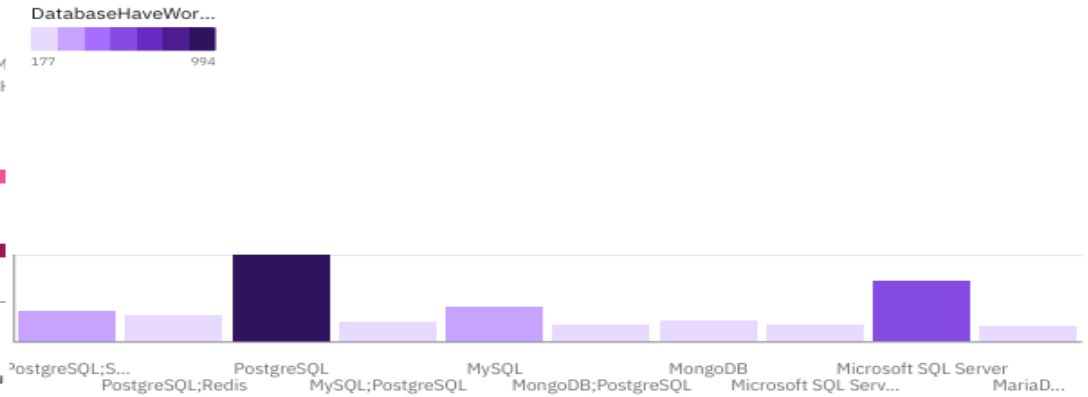
DASHBOARD TAB 1(IBM Cognos Analytics)

Current Technology Usage

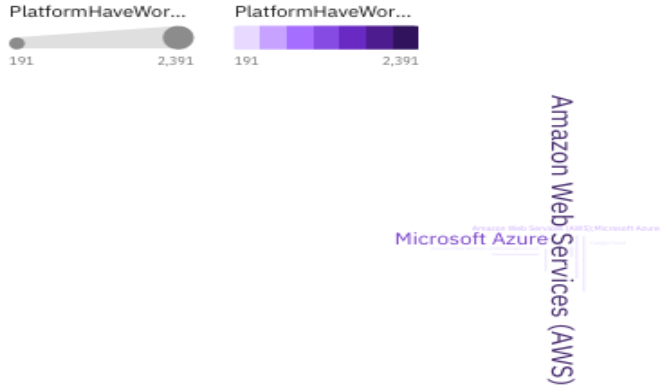
Top 10 Languages Developers Have Worked With



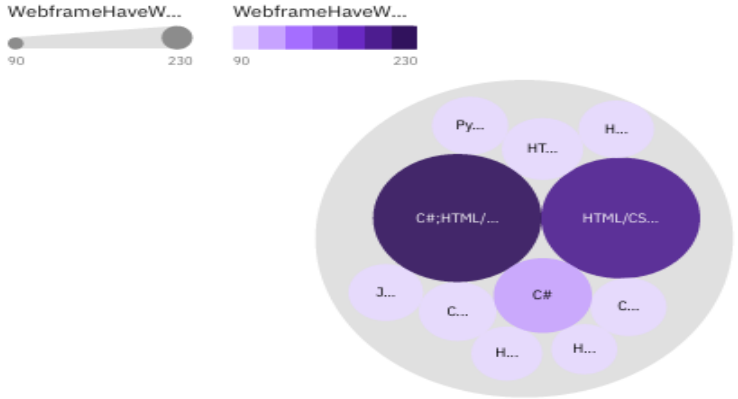
Top 10 Databases Developers Have Worked With



Top 10 Platforms Developers Have Worked With

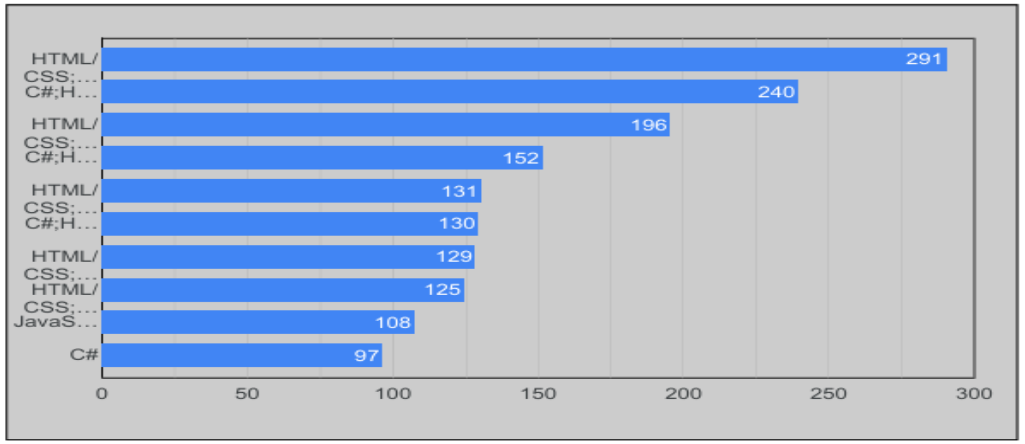


Top 10 Webframe Developers Have Worked With

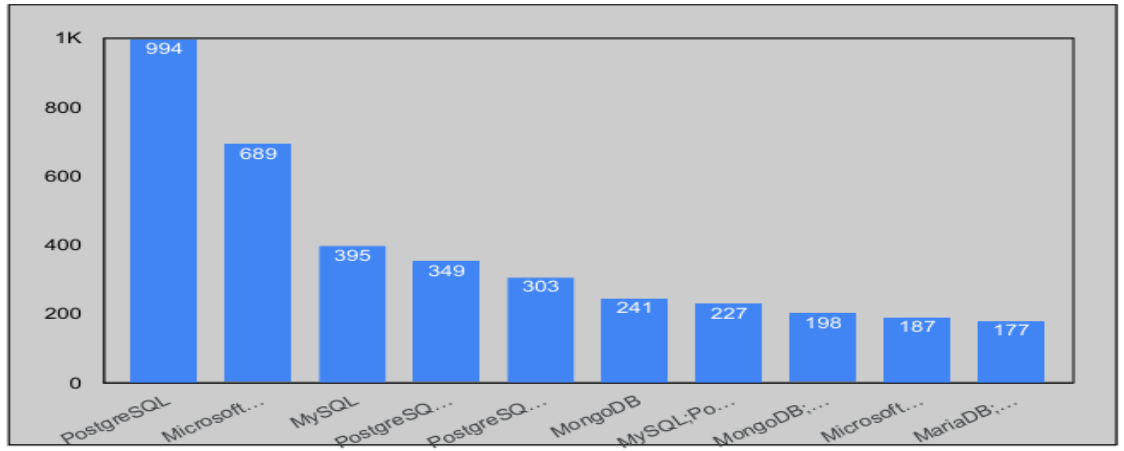


DASHBOARD TAB 1 (Google Looker Studio)

Top 10 Languages used

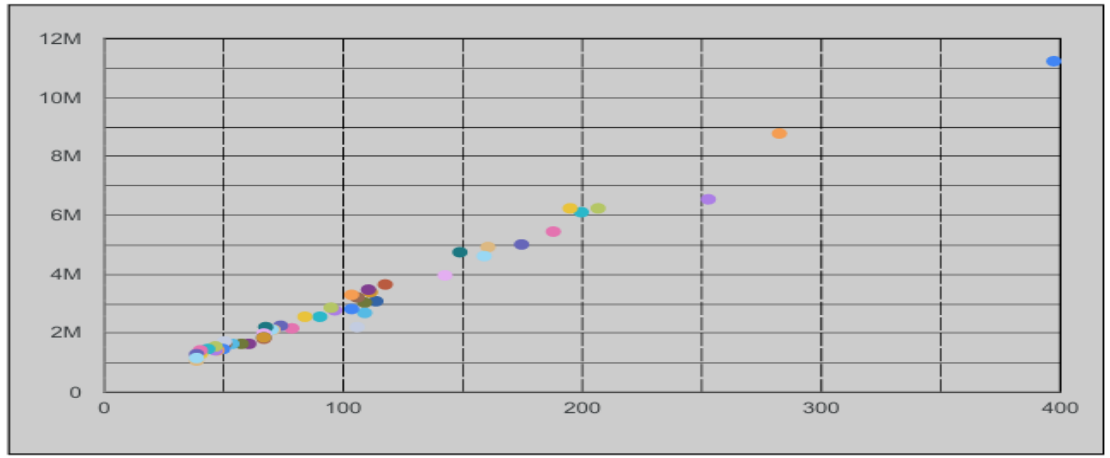


Top 10 Databases used



Top 10 Platforms Used

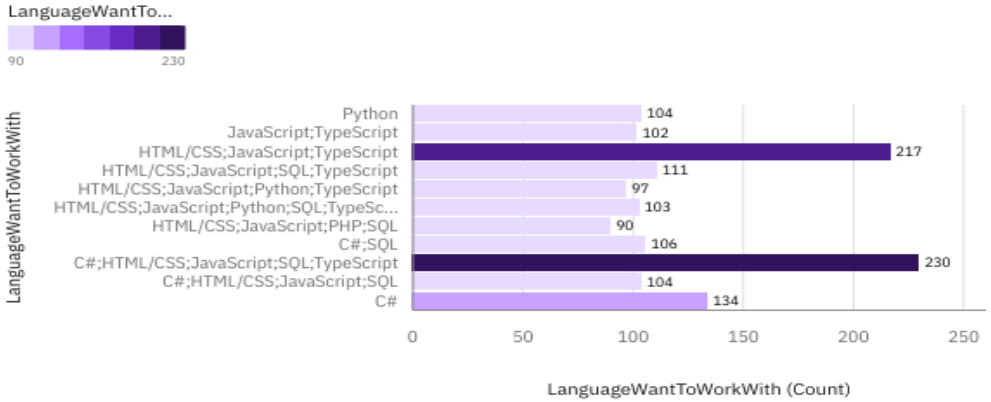
Top 10 Webframes used



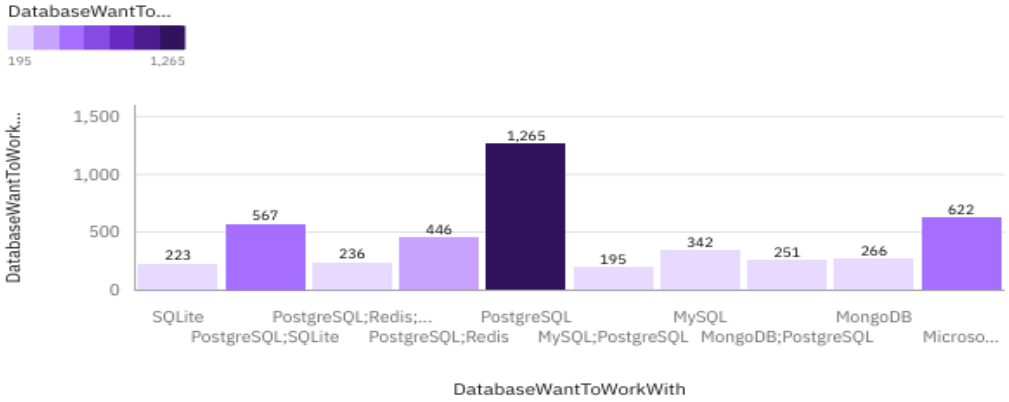
DASHBOARD TAB 2 (IBM Cognos Analytics)

Future Technology Trend

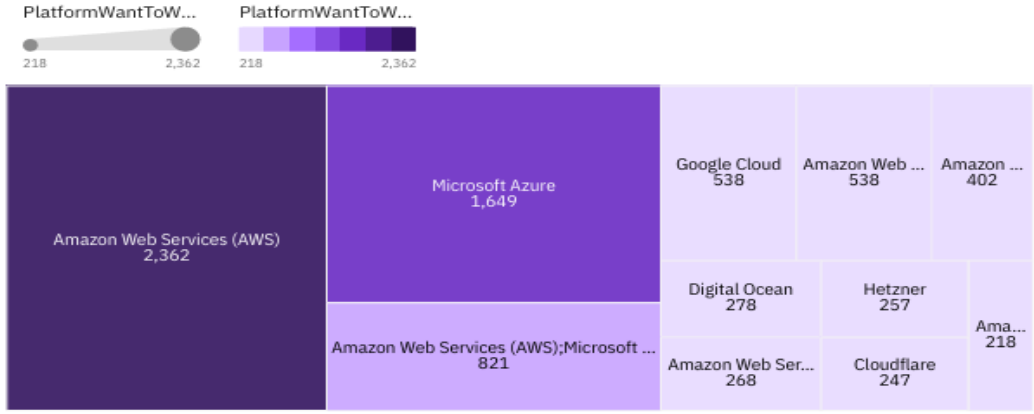
Top 10 Languages Developer Wants To Work With



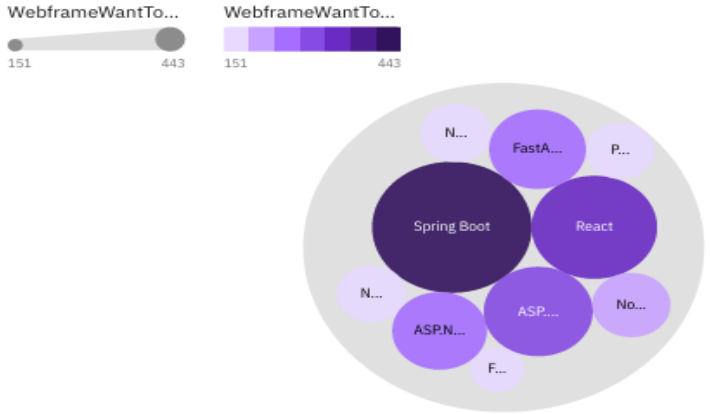
Top 10 Databases Developer Wants To Work With



Top 10 Platforms Developer Wants To Work With

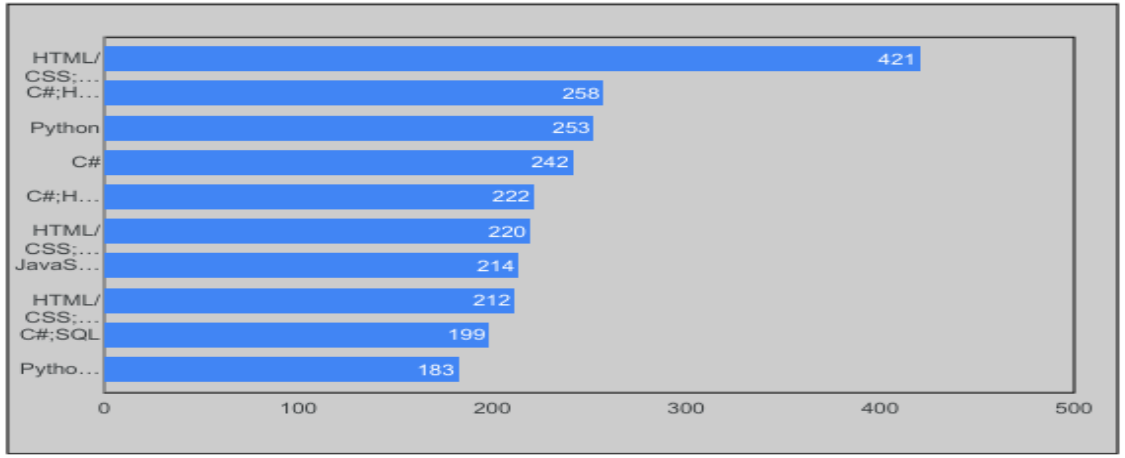


Top 10 Webframe Developer Wants To Work With

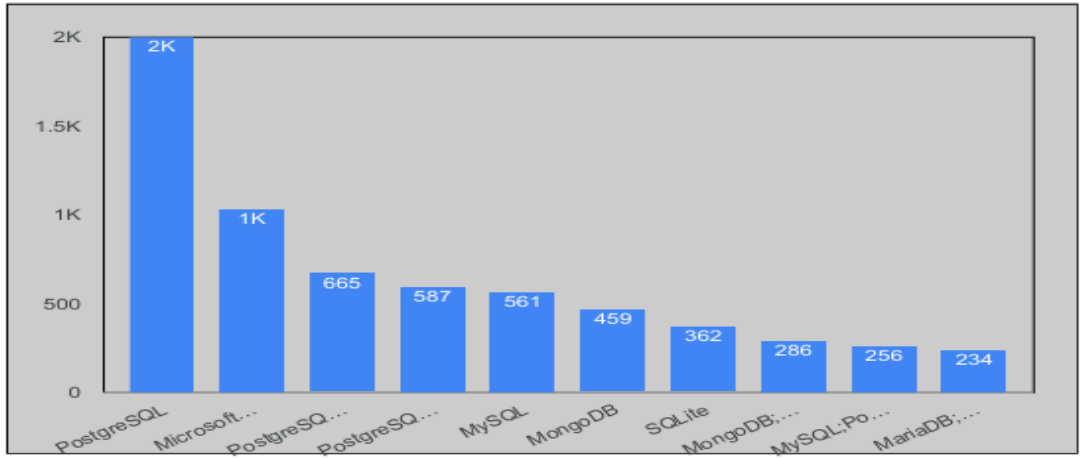


DASHBOARD TAB 2 (Google Looker Studio)

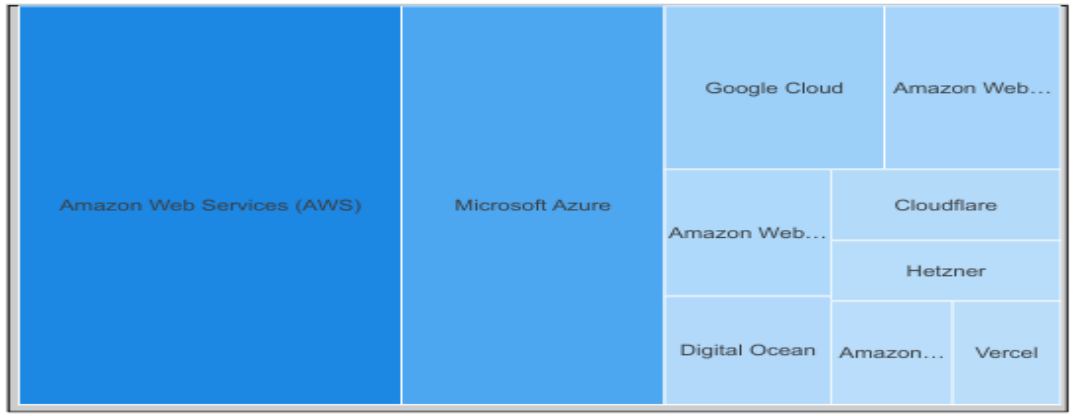
Top 10 Languages Desired Next Year



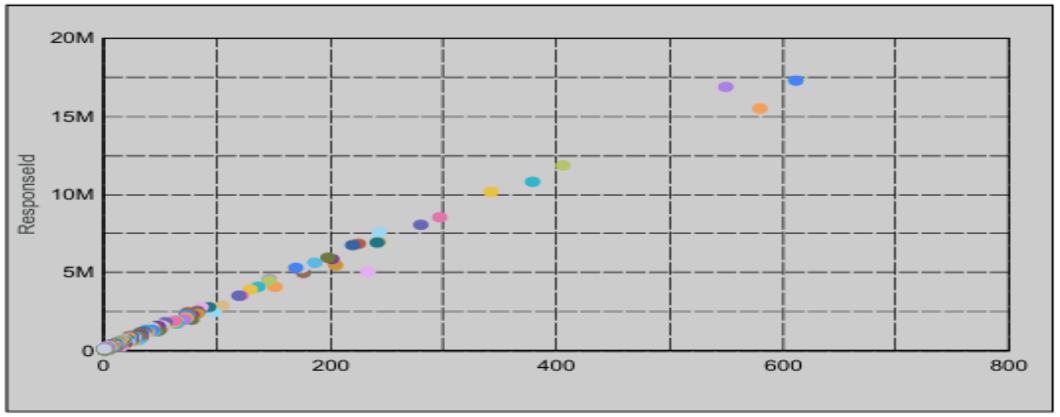
Top 10 Databases Desired Next Year



Top 10 Desired Platforms



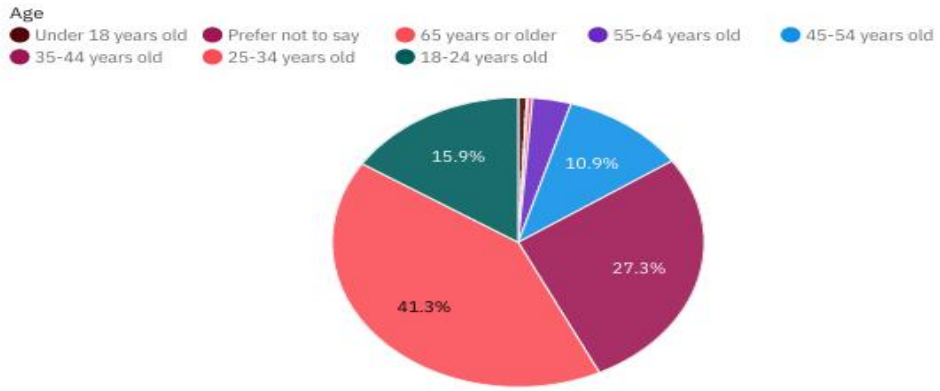
Top 10 Desired Web Frameworks



DASHBOARD TAB 3 (IBM Cognos Analytics)

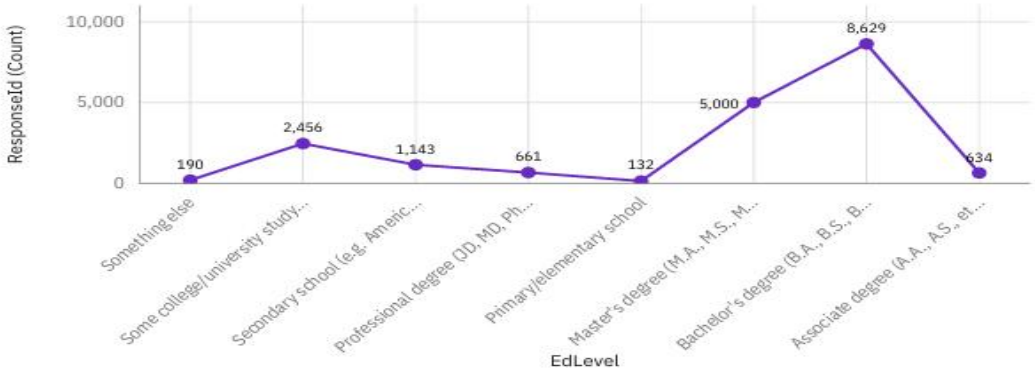
Demographics

Respondent Dsitribution by Age

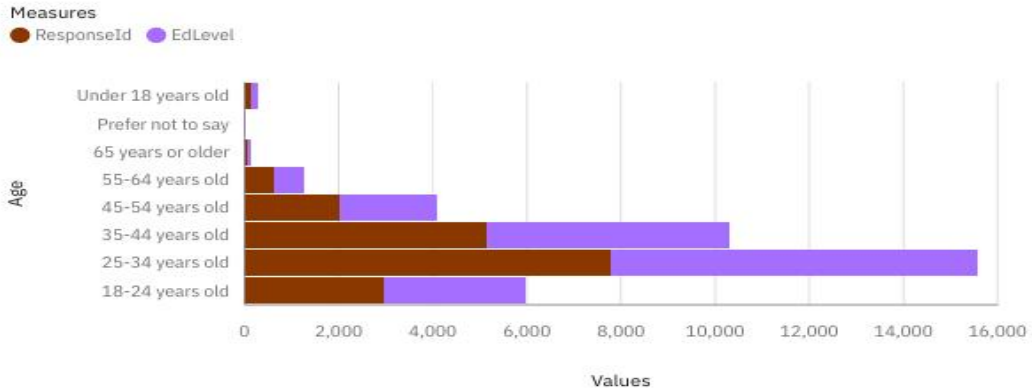


Respondent Count by Country

Respondent distribution by Formal Education Level

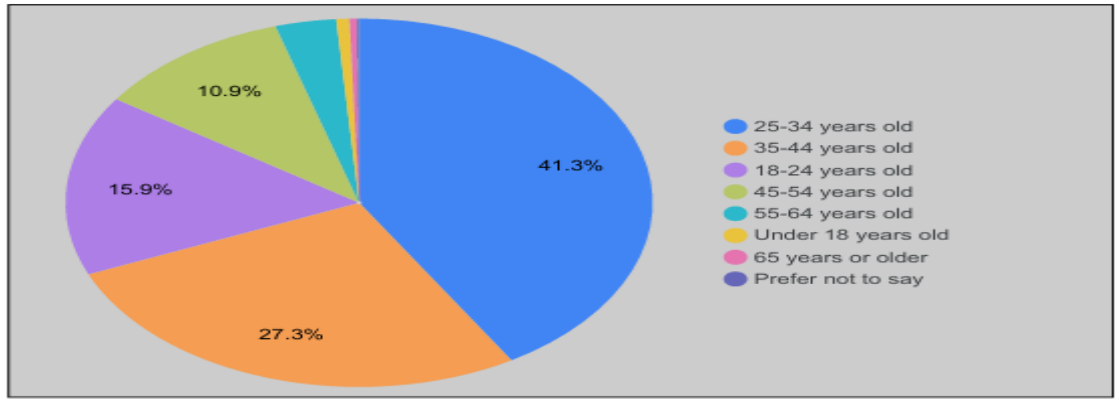


Respondent Count by Age, classified by Education Level

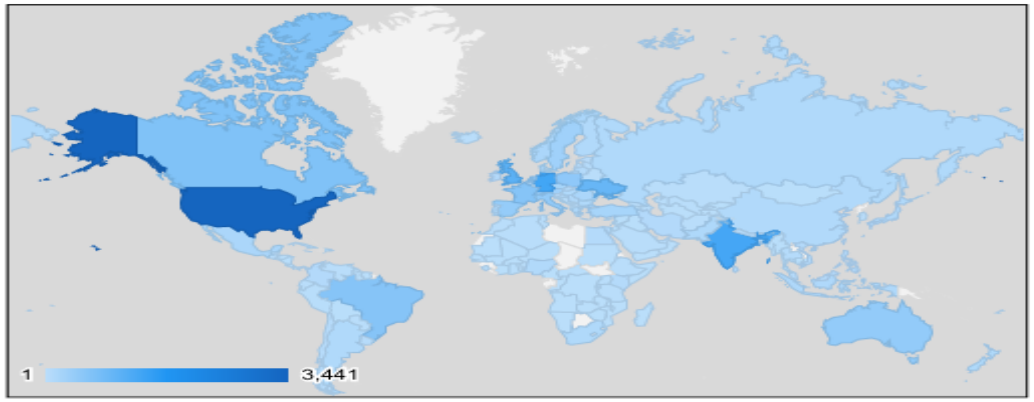


DASHBOARD TAB 3 (Google Looker Studio)

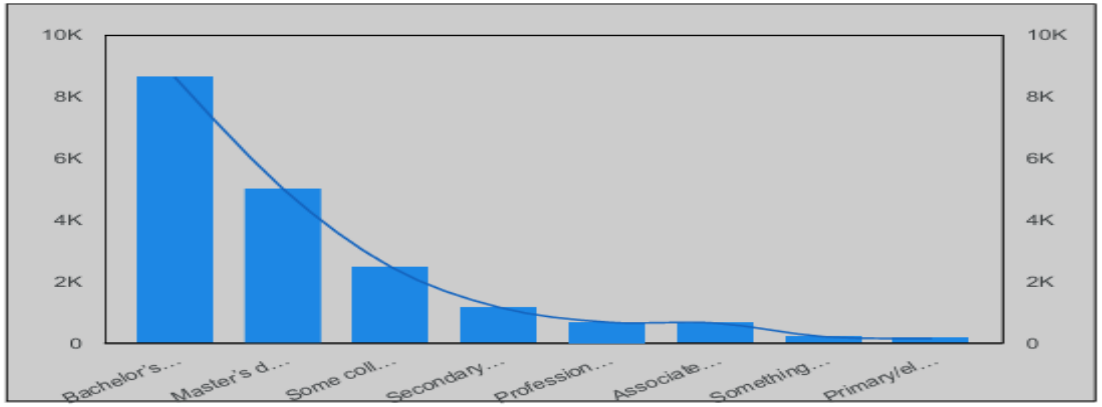
Respondents by Age



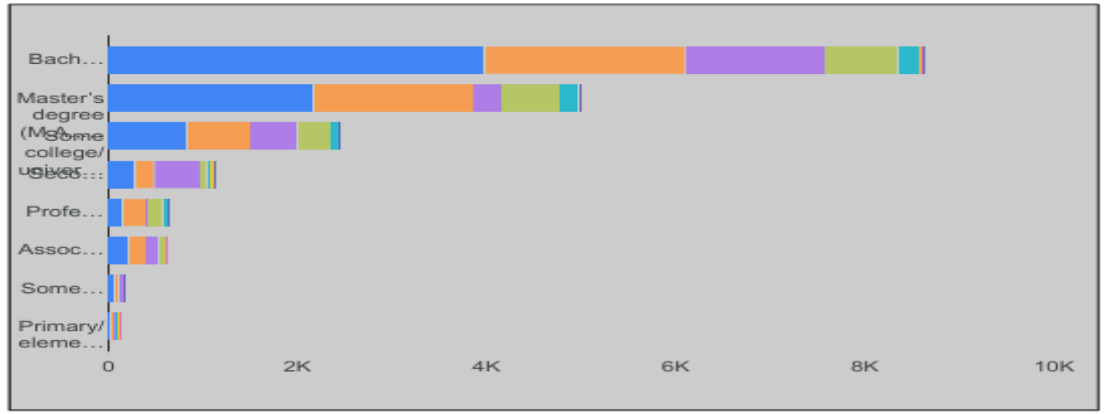
Respondent Count by Country



Respondent Distribution by Education Level



Respondent Count by Age, Classified by Education Level



DISCUSSION

- **Top Technologies Used:** C#, HTML/CSS, JavaScript, PostgreSQL, and AWS are the most commonly used technologies.
- **Future Preferences:** Developers show strong interest in Python, PostgreSQL, React, and cloud platforms like AWS and Azure.
- **Trends:** Full-stack skills and cloud-native tools are in high demand.
- **Demographics:** Majority of respondents are aged 25–34 and hold a Bachelor's or Master’s degree, indicating a young, educated developer base.

OVERALL FINDINGS & IMPLICATIONS

Findings

- C#, HTML/CSS, JavaScript, and PostgreSQL are the most used technologies.
- Python and React are the most desired tools for future development.
- Cloud platforms like AWS, Azure, and Google Cloud dominate both current and future preferences.
- The 25–34 age group is the largest segment, with most holding Bachelor’s or Master’s degrees.

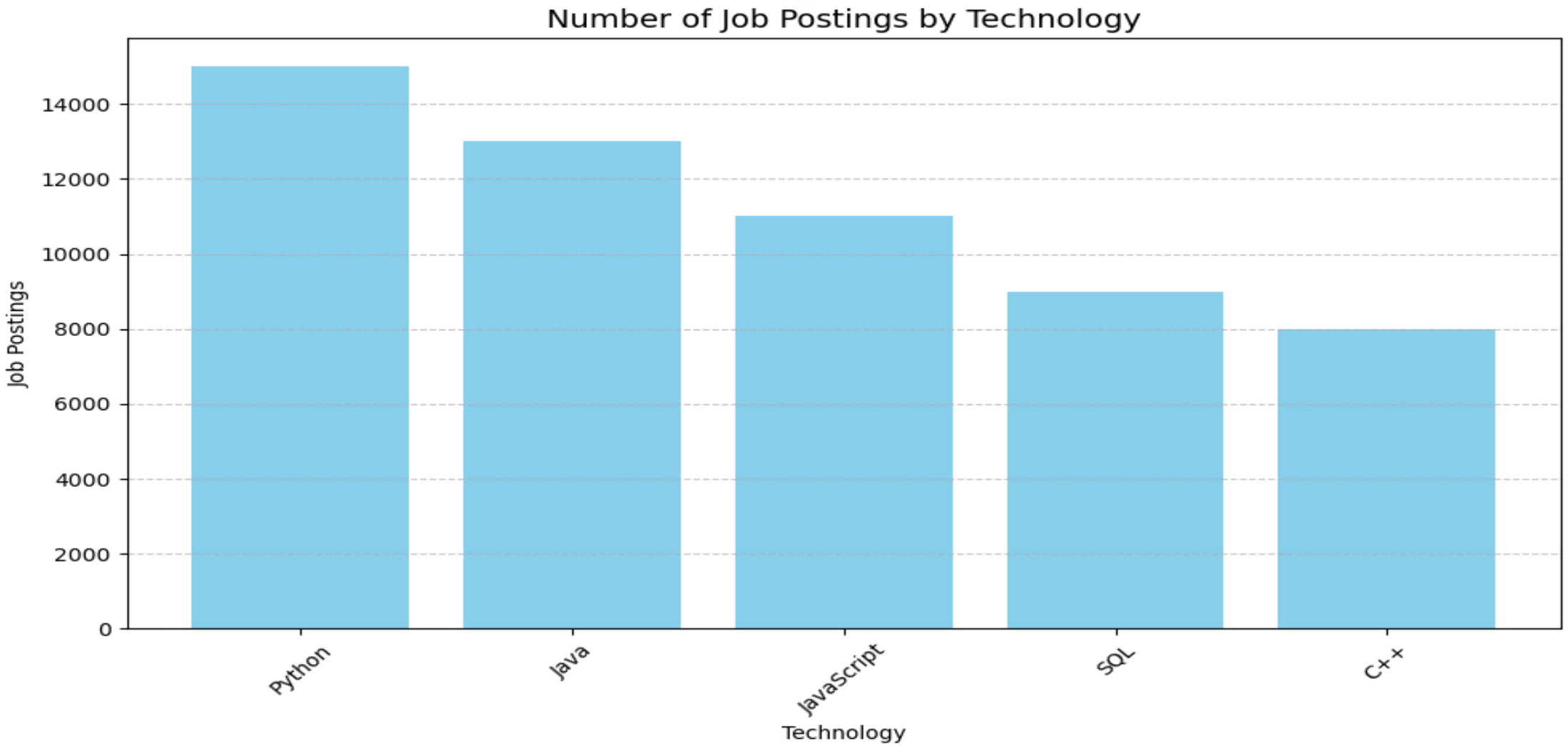
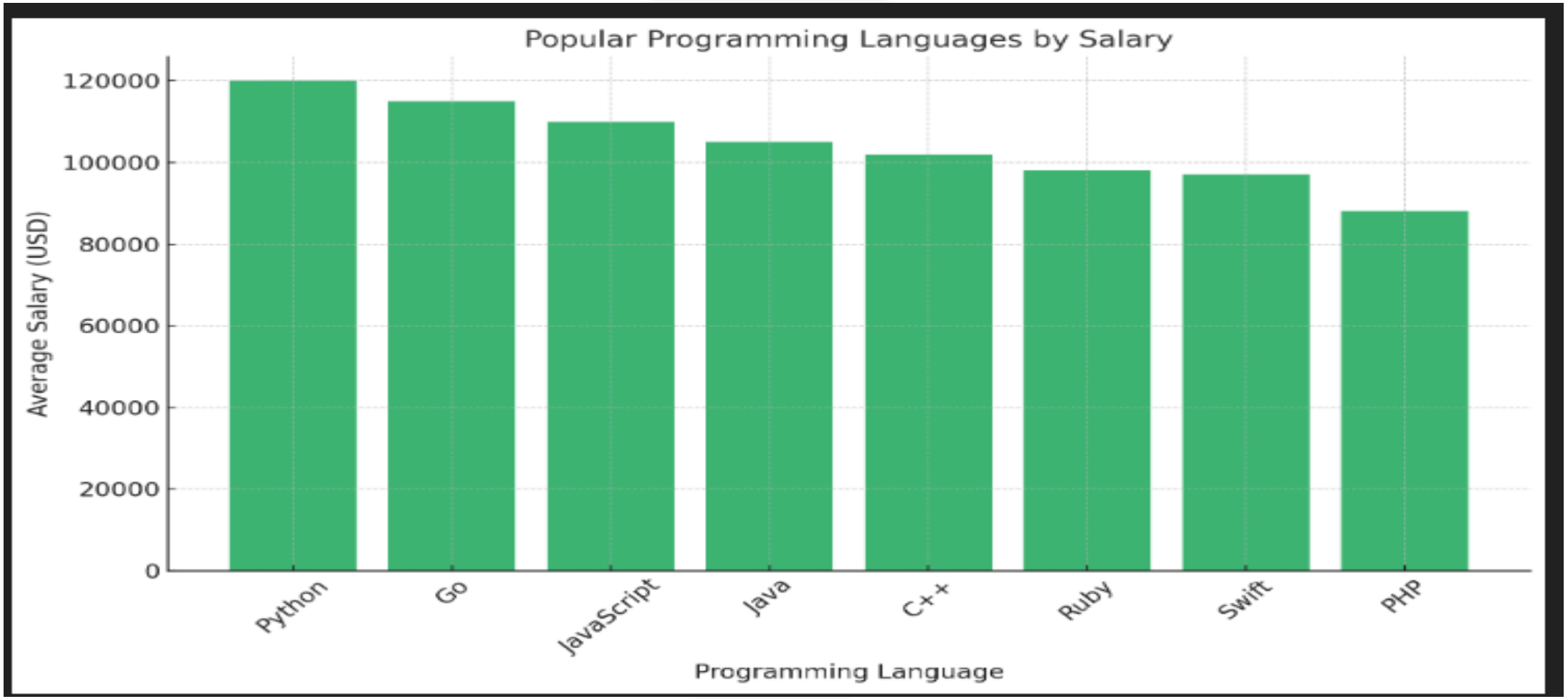
Implications

- Upskilling in Python, PostgreSQL, and React can align with market demand.
- Cloud certifications (e.g., AWS, Azure) will be increasingly valuable.
- Training and recruitment should target young professionals with full-stack and cloud expertise.
- Education programs should focus on modern web frameworks and DevOps/cloud integration.

CONCLUSION

- Python, C#, and JavaScript are among the most used and desired programming languages, indicating strong relevance across domains.
- PostgreSQL has become the dominant database, with growing interest in MongoDB and real-time solutions like Redis.
- AWS remains the leading cloud platform, with rising interest in Azure and Google Cloud, reflecting a cloud-first development shift.
- Developers are increasingly seeking full-stack skills and cloud-native tools, making these key areas for training and hiring.
- The tech workforce is predominantly young (25–34) and highly educated, highlighting a dynamic and adaptable talent pool.

JOB POSTINGS



POPULAR LANGUAGES